

Unit 5: Numerical Methods

Prepared by Staples Education

Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

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Unit 5 at a Glance: Core Sub-Topics

When an initial value problem $\frac{dy}{dt} = f(t, y)$, $y(t_0) = y_0$ cannot be solved in closed form, we march the solution forward in small steps of size h . Every method in this unit is a recipe for turning the slope field f into the next point; they differ only in how many slopes they sample per step and how they weight them.

- **Euler’s Method** — one tangent-line step $y_{n+1} = y_n + hf(t_n, y_n)$; first order, global error $O(h)$.
- **Improved and Modified Euler** — two-slope averaging and the midpoint step; both second order, $O(h^2)$.
- **Heun’s Method** — the predictor–corrector reading of improved Euler as a trapezoidal average of endpoint slopes.
- **The RK4 Method** — four weighted stage slopes $\frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$; fourth order, $O(h^4)$.
- **The Runge–Kutta Family** — the general weighted-stage template that contains Euler, Heun, and RK4 as members.
- **Adaptive Step Size and Comparison** — estimating local error to resize h , and ranking the methods by order of accuracy.

Part I — Condensed Cheat Sheet

5.1 Euler's Method

Euler's method is the simplest numerical solver: from the current point it walks a single step along the tangent line whose slope is $f(t_n, y_n)$. It is the foundation every richer method refines.

The Euler Update

$$y_{n+1} = y_n + h f(t_n, y_n), \quad t_{n+1} = t_n + h.$$

- Each step follows the **tangent line** with slope $f(t_n, y_n)$.
- **Local** truncation error is $O(h^2)$ (the dropped $\frac{h^2}{2}y''$ term); **global** error is $O(h)$, so Euler is a **first-order** method.
- Halving h roughly **halves** the global error.

Worked Example 5.1: One Euler step for $y' = y$

With $y(0) = 1$ and $h = 0.1$, the slope at the start is $f(0, 1) = 1$:

$$y_1 = 1 + 0.1 \cdot 1 = 1.1, \quad t_1 = 0.1.$$

The exact value is $e^{0.1} \approx 1.10517$, so a single first-order step is already low by about 0.005.

5.2 Improved and Modified Euler Methods

Both upgrades sample a *second* slope to capture the curvature that plain Euler ignores, lifting the order from one to two. The improved method averages the slopes at the two endpoints; the modified (midpoint) method reads one slope at the interval's center.

Improved Euler (Endpoint Average) and Modified Euler (Midpoint)

Improved Euler:

$$k_1 = f(t_n, y_n), \quad k_2 = f(t_n + h, y_n + h k_1),$$
$$y_{n+1} = y_n + \frac{h}{2} (k_1 + k_2).$$

Modified Euler (midpoint):

$$k_1 = f(t_n, y_n), \quad k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_1\right),$$
$$y_{n+1} = y_n + h k_2.$$

Both are **second order**, with global error $O(h^2)$: halving h cuts the error by about a factor of four.

Worked Example 5.2: Improved Euler for $y' = y$

With $y(0) = 1$, $h = 0.1$: the predictor slope is $k_1 = 1$ and the corrected slope is $k_2 = f(0.1, 1.1) = 1.1$. Averaging,

$$y_1 = 1 + \frac{0.1}{2} (1 + 1.1) = 1 + 0.05 \cdot 2.1 = 1.105.$$

This already matches $e^{0.1} \approx 1.10517$ to three decimals — far better than the plain Euler 1.1.

5.3 Heun's Method

Heun's method is improved Euler told as a *predictor-corrector*: take a provisional Euler step to predict the endpoint, then correct with the *average* of the slopes at the two ends. That average is exactly the trapezoidal rule applied to $\int f dt$.

Heun's Predictor-Corrector

1. **Predict** the endpoint with a plain Euler step:

$$\tilde{y}_{n+1} = y_n + h f(t_n, y_n).$$

2. **Correct** with the trapezoidal average of the two slopes:

$$y_{n+1} = y_n + \frac{h}{2} \left[f(t_n, y_n) + f(t_{n+1}, \tilde{y}_{n+1}) \right].$$

Averaging the two endpoint slopes recovers the curvature term plain Euler drops, making Heun a **second-order** method — identical to improved Euler.

Common Pitfall: Unequal vs. Equal Slope Weights

The improved Euler / Heun corrector weights the two slopes *equally*, $\frac{h}{2}(k_1 + k_2)$. Do not borrow RK4's $\frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ weights for a two-slope method, and do not forget the predictor uses the *full* step h while the midpoint method uses a *half* step $\frac{h}{2}$.

5.4 The RK4 Method

The classical fourth-order Runge-Kutta method samples four slopes per step — at the start, twice at the midpoint, and once at the end — and blends them with Simpson-like weights 1, 2, 2, 1. It is the workhorse of practical ODE solving.

The Classical RK4 Step

$$\begin{aligned}k_1 &= f(t_n, y_n), \\k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \quad k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \\k_4 &= f(t_n + h, y_n + hk_3), \\y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).\end{aligned}$$

RK4 is **fourth order**, with global error $O(h^4)$: halving h shrinks the error by about a factor of **16**.

Worked Example 5.4: One RK4 step for $y' = y$

With $y(0) = 1$, $h = 0.1$:

$$\begin{aligned}k_1 &= 1, \quad k_2 = 1.05, \quad k_3 = 1.0525, \quad k_4 = 1.10525. \\y_1 &= 1 + \frac{0.1}{6}(1 + 2(1.05) + 2(1.0525) + 1.10525) = 1.105171.\end{aligned}$$

Against $e^{0.1} = 1.1051709\dots$ the error is below 10^{-6} in a single step.

5.5 The Runge–Kutta Family

Euler, the midpoint method, Heun, and RK4 are all instances of one template: take a weighted average of slopes sampled at chosen points across the step. The *number of stages* and the *weights* fix the order of accuracy.

The General Explicit Runge–Kutta Step

For s stages with weights b_i , nodes c_i , and coupling coefficients a_{ij} :

$$k_i = f\left(t_n + c_i h, y_n + h \sum_{j < i} a_{ij} k_j\right), \quad y_{n+1} = y_n + h \sum_{i=1}^s b_i k_i.$$

- Consistency requires $\sum_i b_i = 1$ (the weights sum to one).
- Euler is the 1-stage member; improved Euler, the midpoint method, and Heun are 2-stage second-order members; RK4 is the classic 4-stage fourth-order member.

5.6 Adaptive Step Size and Method Comparison

Rather than fix h in advance, adaptive solvers *estimate* the local error each step and resize h to hold it under a tolerance — small steps through sharp turns, large steps across calm stretches.

Estimating Local Error to Resize h

1. Advance the step two ways of *different order* (e.g. an embedded RK4/RK5 pair, or one full step vs. two half steps).
2. The difference of the two estimates approximates the **local error**.
3. If the error exceeds the tolerance, **shrink** h and retry; if it is comfortably under, **grow** h for the next step.

Common Pitfall: Confusing Order with Step Count

A method's *order* p is how fast error falls as $h \rightarrow 0$ (global error $O(h^p)$), not how many steps it takes. RK4 with a large h can beat Euler with a tiny h at equal cost. And smaller h is not free: too small, and accumulated roundoff begins to swamp the truncation gain.

Unit 5 Key Takeaway

Every solver here marches $y_{n+1} = y_n + h$ (weighted slope); they differ only in the slope. **Euler** samples one slope (first order, $O(h)$); **improved/modified Euler and Heun** average two (second order, $O(h^2)$); **RK4** blends four with weights **1, 2, 2, 1** (fourth order, $O(h^4)$). All sit inside the **Runge–Kutta family**, and **adaptive stepping** tunes h from an on-the-fly local-error estimate.

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from a single Euler step, through the two-slope second-order methods (improved Euler, modified Euler, and Heun), into the four-stage RK4 step, and close with a method comparison and an adaptive-stepping capstone. Throughout, take $\frac{dy}{dt} = f(t, y)$ with step size h . Solutions follow in Part III.

1. Write the Euler update formula, then take one Euler step for $\frac{dy}{dt} = t + y$ with $y(0) = 1$ and $h = 0.2$. Report y_1 and t_1 .
2. Continue from Problem 1 and take a *second* Euler step to find y_2 at $t_2 = 0.4$.
3. State the local and global truncation error orders of Euler's method, and the factor by which the global error changes when h is halved.
4. Take one *improved Euler* step for $\frac{dy}{dt} = t + y$ with $y(0) = 1$ and $h = 0.2$. Show k_1 , k_2 , and y_1 .
5. Take one *modified Euler* (midpoint) step for the same problem, $\frac{dy}{dt} = t + y$, $y(0) = 1$, $h = 0.2$. Show k_1 , k_2 , and y_1 .
6. Write Heun's predictor–corrector formulas and explain in one line why the method is second order.
7. For a general $f(t, y)$, write the four RK4 stage slopes k_1, k_2, k_3, k_4 and the final RK4 update for y_{n+1} .
8. Take one RK4 step for $\frac{dy}{dt} = y$ with $y(0) = 1$ and $h = 0.1$. Compute y_1 and compare it to $e^{0.1}$.
9. Rank Euler, improved/modified Euler, Heun, and RK4 by order of accuracy, and give the factor by which each method's global error drops when h is halved.
10. Explain how an adaptive solver uses two estimates of *different* order to decide whether to shrink or grow the step size h .
11. **(Capstone)** Take one RK4 step for $\frac{dy}{dt} = t + y$ with $y(0) = 1$ and $h = 0.2$. Compute y_1 , then compare it against the Euler result (Problem 1) and the improved Euler result (Problem 4) using the exact solution $y = 2e^t - t - 1$.

Part III — Complete Solutions

Solution 1. The Euler update is $y_{n+1} = y_n + h f(t_n, y_n)$. With $f(t, y) = t + y$, the starting slope is $f(0, 1) = 0 + 1 = 1$:

$$y_1 = 1 + 0.2 \cdot 1 = 1.2, \quad t_1 = 0.2.$$

Solution 2. From $(t_1, y_1) = (0.2, 1.2)$ the slope is $f(0.2, 1.2) = 0.2 + 1.2 = 1.4$:

$$y_2 = 1.2 + 0.2 \cdot 1.4 = 1.48, \quad t_2 = 0.4.$$

Solution 3. A single Euler step drops the $\frac{h^2}{2}y''$ term, so the **local** truncation error is $O(h^2)$. Accumulated over $\sim 1/h$ steps, the **global** error is $O(h)$, making Euler a first-order method. Halving h therefore

cuts the global error by about a factor of 2.

Solution 4. Improved Euler uses $k_1 = f(t_n, y_n)$ and $k_2 = f(t_n + h, y_n + hk_1)$, then averages:

$$k_1 = f(0, 1) = 1, \quad k_2 = f(0.2, 1 + 0.2 \cdot 1) = f(0.2, 1.2) = 1.4.$$

$$y_1 = 1 + \frac{0.2}{2} (1 + 1.4) = 1 + 0.1 \cdot 2.4 = 1.24.$$

Solution 5. Modified Euler reads the slope at the midpoint after a half-step prediction:

$$k_1 = f(0, 1) = 1, \quad k_2 = f(0.1, 1 + 0.1 \cdot 1) = f(0.1, 1.1) = 1.2.$$

$$y_1 = 1 + h k_2 = 1 + 0.2 \cdot 1.2 = 1.24.$$

Here both second-order methods agree to the digits shown.

Solution 6. Heun predicts with a full Euler step and corrects with the trapezoidal average of the endpoint slopes:

$$\tilde{y}_{n+1} = y_n + h f(t_n, y_n), \quad y_{n+1} = y_n + \frac{h}{2} [f(t_n, y_n) + f(t_{n+1}, \tilde{y}_{n+1})].$$

Averaging the two endpoint slopes restores the leading curvature term that plain Euler discards, so the global error is $O(h^2)$ — second order.

Solution 7. The classical RK4 stages and update are

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \quad k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \\ k_4 &= f(t_n + h, y_n + hk_3), \\ y_{n+1} &= y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

Solution 8. For $f(t, y) = y$ with $y(0) = 1$, $h = 0.1$:

$$k_1 = 1, \quad k_2 = 1 + 0.05 \cdot 1 = 1.05, \quad k_3 = 1 + 0.05 \cdot 1.05 = 1.0525,$$

$$k_4 = 1 + 0.1 \cdot 1.0525 = 1.10525.$$

$$y_1 = 1 + \frac{0.1}{6}(1 + 2(1.05) + 2(1.0525) + 1.10525) = 1 + \frac{0.1}{6}(6.31025) = 1.105171.$$

Against $e^{0.1} = 1.1051709\dots$ the error is below 10^{-6} .

Solution 9. Ranked by order of accuracy:

Euler	1st order $O(h)$	halving $h \Rightarrow \div 2$
Improved / Modified Euler	2nd order $O(h^2)$	halving $h \Rightarrow \div 4$
Heun	2nd order $O(h^2)$	halving $h \Rightarrow \div 4$
RK4	4th order $O(h^4)$	halving $h \Rightarrow \div 16$

Solution 10. An adaptive solver advances each step two ways of *different* order — for example an embedded RK4/RK5 pair, or one full step compared against two half steps. The difference between the two estimates approximates the local error. If that estimate **exceeds** the tolerance, the solver **shrinks** h and retries the step; if it sits comfortably **under** the tolerance, the solver **grows** h for the next step, so the work concentrates where the solution changes fastest.

Solution (Capstone). RK4 for $f(t, y) = t + y$, $y(0) = 1$, $h = 0.2$:

$$k_1 = f(0, 1) = 1, \quad k_2 = f(0.1, 1 + 0.1 \cdot 1) = f(0.1, 1.1) = 1.2,$$

$$k_3 = f(0.1, 1 + 0.1 \cdot 1.2) = f(0.1, 1.12) = 1.22, \quad k_4 = f(0.2, 1 + 0.2 \cdot 1.22) = f(0.2, 1.244) = 1.444.$$

$$y_1 = 1 + \frac{0.2}{6}(1 + 2(1.2) + 2(1.22) + 1.444) = 1 + \frac{0.2}{6}(7.284) = 1.242805.$$

The exact solution gives $y(0.2) = 2e^{0.2} - 0.2 - 1 = 1.242806$, so RK4 is correct to six figures. Compare the one-step estimates:

Euler 1.2 (err $\approx 4.3 \times 10^{-2}$), Improved 1.24 (err $\approx 2.8 \times 10^{-3}$), RK4 1.242805 (err $\approx 10^{-6}$).

Unit 5 Key Takeaway

Accuracy is bought with slopes per step. Euler uses one (first order); improved/modified Euler and Heun average two (second order); RK4 weights four as $\frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ (fourth order). On $y' = t + y$ over a single $h = 0.2$ step, the errors fall from $\sim 10^{-2}$ to $\sim 10^{-6}$ across the ladder — and adaptive stepping spends that accuracy budget only where the solution demands it.